

MODELING AND FORECASTING RAINFALL PATTERNS IN RATNAPURA, SRI LANKA USING NAR AND ARIMA(SARIMA) MODELS

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Accurate forecasting of tropical rainfall is essential for disaster risk reduction in flood-prone regions, particularly as climate change increases the frequency and intensity of extreme weather events. This study investigates rainfall forecasting methodologies for Ratnapura, Sri Lanka, a region highly vulnerable to flooding, by comparing the performance of traditional stochastic time-series models with artificial neural network approaches.

A comprehensive rainfall dataset covering the period from 2000 to 2025 was preprocessed using R, including chronological ordering, linear interpolation for missing-value imputation, and outlier treatment to manage the high variability characteristic of tropical monsoon climates. Time-series forecasting models, including Automated ARIMA, Manual ARIMA, and Seasonal ARIMA (SARIMA), were developed and evaluated. In parallel, Nonlinear Autoregressive (NAR) neural networks with various architectures were implemented, including dynamically optimized configurations, manually tuned hidden layers, and BoxCox variance stabilization transformations. Model performance was assessed using standard forecasting accuracy metrics and residual diagnostic tests. The results demonstrated that the manually configured SARIMA model, ARIMA(4,2,2)(0,1,1)[12], achieved the highest out-of-sample forecasting accuracy of 66.79%, with residual analysis confirming statistically independent and unbiased errors (Ljung-Box, $*p > 0.05$). In contrast, the best-performing NAR neural network achieved an accuracy of 60.70%, while variance stabilization techniques further reduced predictive performance.

The findings indicate that carefully calibrated stochastic time-series models provide a more robust, interpretable, and reliable framework for tropical rainfall forecasting than the neural network architectures examined in this study. These results highlight the continued relevance of traditional statistical approaches for supporting early warning systems and disaster risk management in tropical regions.

Keywords: Tropical Rainfall Forecasting; Time Series Analysis; SARIMA Model; Artificial Neural Networks; Disaster Risk Reduction.